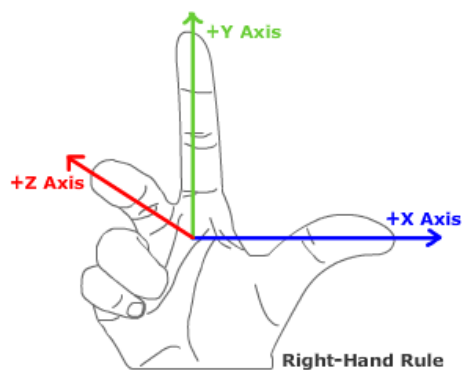


07. Kinematics, inverse kinematics, Programming of a simulated robotic arm

Repetition

3D transformations

- **Position:** 3 element offset vector



- **Orientation:** 3 x 3 rotation matrix
 - additional orientation representations: Euler angles, RPY, angle axis, quaternion
- **Pose** (pose): 4 x 4 transformation matrix
- **Coordinate system** (frame): zero point, 3 axis, 3 base vector, right-hand rule
- **Homogeneous transformations:** rotation and translation together
 - e.g. $\mathbf{R}$ for rotation and $\mathbf{v}$ for translation: